ARTIFICIAL INTELLIGENCE (CT 653)

Tribhuvan University | Institute of Engineering | BCT | Year III / Part II

CHAPTER 1: Introduction to Artificial Intelligence

Syllabus Hours: 4 hrs | Marks Weight: 7 marks

Complete Exam-Ready Answers | All Past Questions Covered (2070–2082)

CHAPTER 1 SYLLABUS TOPICS

- 1.1 Definition of Artificial Intelligence
- 1.2 Importance of Artificial Intelligence
- 1.3 AI and Related Fields
- 1.4 Brief History of Artificial Intelligence
- 1.5 Applications of Artificial Intelligence
- 1.6 Definition and Importance of Knowledge and Learning

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SECTION 1: Definition, Types & Importance of Artificial Intelligence

Exam Ref: 2082 Kartik, 2081 Chaitra, 2080 Ashwin Back, 2079 Chaitra, 2078 Chaitra, 2077 Chaitra, 2076 Baisakh, 2075 Baisakh, 2074 Bhadra, 2073 Bhadra, 2072 Ashwin, 2071 Bhadra, 2070 Magh

Q1. Define Artificial Intelligence (AI). Explain different types of Intelligent Agents with examples. [2+6] (2080 Ashwin Back, 2082 Kartik)

Answer: Definition of Artificial Intelligence

Artificial Intelligence (AI) is the branch of computer science that deals with the simulation of intelligent behavior in computers. It is the science and engineering of making machines that can perform tasks that would normally require human intelligence.

Russell & Norvig Definition: AI is the study of agents that perceive their environment and take actions that maximize their chances of success.

AI can be classified along two dimensions:

Dimension	Human-Based	Rationality-Based
Thinking	Think Like Humans (Cognitive Science)	Think Rationally (Logic & Reasoning)
Acting	Act Like Humans (Turing Test)	Act Rationally (Rational Agents)

Types of Intelligent Agents:

Agent: An entity that perceives its environment through sensors and acts upon it through actuators.

PEAS Framework: Performance measure, Environment, Actuators, Sensors — used to describe any agent.

Agent Type	Description	Characteristics	Example
Simple Reflex Agent	Acts only on current percept; ignores history	Condition-Action rules; no memory	Thermostat, vacuum cleaner
Model-Based Reflex Agent	Maintains internal state/model of world	Handles partial observability; has memory	Robot with map memory
Goal-Based Agent	Chooses actions that achieve a goal	Searches for action sequences to reach goal	Navigation AI, chess player
Utility-Based Agent	Maximizes a utility function (happiness measure)	Handles conflicting goals; finds best action	Self-driving car, stock trader
Learning Agent	Improves performance over time from experience	Has learning element, performance element, critic	AlphaGo, spam filter

PEAS Description Examples:

Agent	Performance Measure	Environment	Actuators	Sensors

Automated Robot (Factory)	Parts assembled correctly, speed, no defects	Factory floor, conveyor belt, parts	Robotic arms, welders	Cameras, touch sensors, position sensors
Taxi Driver	Safe trip, speed, destination reached	Roads, traffic, passengers	Steering, brake, accelerator	GPS, cameras, speedometer
Medical Diagnosis	Correct diagnosis, patient health	Hospital, patient symptoms, test results	Display, prescription printer	Keyboard input, medical sensors
Spam Filter	Correct classification of emails	Email inbox	Mark/delete email	Email content, headers

Q2. What is Artificial Intelligence? Explain AI applications in different areas as a problem solving tool. [Most Repeated] **[2+4]** (2081 Chaitra, 2078 Chaitra, 2076 Baisakh)

Answer:

AI enables computers to perform cognitive tasks like understanding language, recognizing images, making decisions, solving problems, and learning from experience. Key applications:

Domain	AI Application	Example System
Healthcare	Medical diagnosis, drug discovery, robotic surgery	IBM Watson Oncology, Da Vinci robot
Natural Language Processing	Machine translation, chatbots, speech recognition	Google Translate, ChatGPT, Siri
Computer Vision	Image recognition, facial recognition, autonomous vehicles	Tesla Autopilot, Face ID
Game Playing	Strategic board games, video game AI	AlphaGo (Go), Stockfish (Chess)
Expert Systems	Knowledge-based decision support	MYCIN (medical), DENDRAL (chemistry)
Robotics	Industrial automation, exploration	Boston Dynamics, Mars Rover
Finance	Fraud detection, algorithmic trading	PayPal fraud detection, HFT systems
Education	Personalized learning, intelligent tutoring	Duolingo, Khan Academy AI
Search Engines	Information retrieval, ranking	Google PageRank, semantic search

Q3. Can machine think? Explain your stance with appropriate reasons. Relate it with Turing Test and Reverse Turing Test. **[2+4]** (2080 Chaitra)

Answer: The Turing Test

Turing Test (Imitation Game): Proposed by Alan Turing (1950) in his paper 'Computing Machinery and Intelligence'. A human judge has a text conversation with both a human and a machine (without seeing either). If the judge cannot reliably tell which is which, the machine is said to have passed the Turing Test — demonstrating human-like intelligence.

Can Machines Think? Turing avoided this philosophical question and replaced it with the operational question: can machines BEHAVE intelligently? His answer: if a machine behaves indistinguishably from a thinking being, we should accept that it thinks.

Arguments FOR machines thinking:

- Machines can learn, recognize patterns, process language, and solve complex problems.
- ELIZA (1966) fooled many users into believing it was human.
- Modern LLMs pass many informal Turing-like tests.

Arguments AGAINST:

- The Chinese Room Argument (Searle): A machine can manipulate symbols without understanding their meaning — syntax ≠ semantics.
- Machines lack consciousness, emotions, and genuine understanding.

Reverse Turing Test (CAPTCHA): A test where a machine distinguishes between humans and other machines. A human judge asks questions; if the machine correctly identifies which responder is human, it passes. This is used in CAPTCHA systems online to block bots.

Total Turing Test: Extends the Turing Test to include visual and physical interaction, requiring Computer Vision and Robotics in addition to NLP.

SECTION 2: History, Importance & Related Fields of AI

Exam Ref: 2079 Chaitra, 2077 Chaitra, 2075 Bhadra, 2073 Magh, 2071 Bhadra, 2070 Magh

Q4. Define Artificial Intelligence (AI). Discuss brief history of AI with Chronological development. Discuss two fields of application of AI. [2+3+3] (2077 Chaitra, 2070 Magh)

Answer: Brief History of AI — Chronological Development

Period	Key Events & Milestones
1943	McCulloch & Pitts publish first mathematical model of a neuron (beginning of neural networks)
1950	Alan Turing publishes 'Computing Machinery and Intelligence'; proposes Turing Test
1956	John McCarthy coins the term 'Artificial Intelligence' at Dartmouth Conference — official birth of AI
1957	First perceptron implemented by Frank Rosenblatt
1966	ELIZA chatbot created by Weizenbaum — first NLP program to simulate conversation
1969	SHAKEY — first mobile robot using AI reasoning developed at Stanford
1972	PROLOG programming language developed
1980s	Expert Systems boom — MYCIN (medical), DENDRAL (chemistry); Japan's Fifth Generation Computer Project
1986	Backpropagation algorithm popularized for neural network training
1997	IBM Deep Blue defeats world chess champion Garry Kasparov
2002	Roomba — first commercially successful AI-powered autonomous robot
2011	IBM Watson wins Jeopardy! against human champions; Apple introduces Siri
2012	AlexNet wins ImageNet — deep learning revolution begins
2016	AlphaGo defeats Go world champion Lee Sedol
2022–Now	Large Language Models (ChatGPT, Claude, Gemini) — AI becomes mainstream

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Q5. What is Artificial Intelligence? Differentiate between Natural (Human) Intelligence and Artificial Intelligence. Can AI replace humans? Justify. [1+3+4] (2079 Chaitra)

Answer: Human Intelligence vs Artificial Intelligence

Feature	Human Intelligence	Artificial Intelligence
Origin	Biological — evolved over millions of years	Engineered — programmed/trained by humans
Learning	Learns from limited data + intuition + experience	Learns from massive datasets (big data)
Speed	Slow — biological neural processing	Very fast — electronic processing

Memory	Limited and can forget	Virtually unlimited, perfect recall
Creativity	High — generates truly novel ideas	Limited — recombines existing patterns
Emotion	Has emotions, motivations, desires	No genuine emotions (only simulated)
Context	Excellent common sense, understands context deeply	Still weak at true common sense reasoning
Energy	~20 watts (very efficient)	Massive — large models use megawatts
Adaptability	Adapts to completely new situations easily	Struggles outside trained domain
Consciousness	Has self-awareness, subjective experience	No consciousness

Can AI Replace Humans?

Partial replacement YES: AI can replace humans in repetitive, data-driven, pattern-recognition tasks — manufacturing, data entry, basic diagnosis, customer support bots.

Full replacement NO: AI cannot replicate human creativity, moral judgment, emotional intelligence, leadership, and general common sense. AI works best as a TOOL that augments human capabilities, not replaces them entirely.

Conclusion: AI will transform jobs and automate many tasks, but human oversight, creativity, and ethical judgment will remain irreplaceable for the foreseeable future.

Q6. What is Artificial Intelligence? Explain AI and related fields. **[2+5]** (2074 Bhadra, 2075 Bhadra)

Answer: AI and Related Fields

Related Field	Relationship to AI	Contribution to AI
Mathematics	AI uses mathematical theories for reasoning, optimization, probability	Logic, set theory, graph theory, statistics, probability
Computer Science	AI is a subfield of CS; uses data structures, algorithms, programming	Algorithms, complexity theory, programming languages
Psychology / Cognitive Science	AI models human thought and behavior	Cognitive architecture, memory models, learning theories
Philosophy	Provides foundations for logic, knowledge, ethics in AI	Epistemology, formal logic, theory of mind
Neuroscience	Biological neural networks inspire artificial neural networks	Neural architecture, learning mechanisms, synaptic models
Linguistics	NLP requires understanding language structure and meaning	Syntax, semantics, pragmatics, grammar

Statistics	Machine learning is heavily statistics-based	Bayesian inference, regression, hypothesis testing
Control Theory	Robotics and agent control use control theory	Feedback systems, PID controllers, stability
Operations Research	Optimization problems in AI planning	Linear programming, scheduling, resource allocation

SECTION 3: Intelligent Agents & PEAS Framework

Exam Ref: 2082 Kartik, 2081 Ashwin, 2079 Chaitra, 2078 Chaitra, 2078 Baishakh, 2075 Bhadra

Q7. What is intelligent agent? Do we need PEAS for designing an intelligent agent? If so, define PEAS and provide the PEAS information for Taxi Driver Agent and Automated Robot in a manufacturing plant. [1+5] (2081 Ashwin)

Answer: Intelligent Agent & PEAS

Intelligent Agent: An intelligent agent is any entity that perceives its environment through SENSORS and acts upon that environment through ACTUATORS to achieve its goals. It is the central concept of modern AI.

Properties of an Intelligent Agent:

- Autonomy: Operates without constant human guidance.
- Reactivity: Responds to changes in the environment in a timely manner.
- Proactivity: Takes initiative to achieve its goals.
- Social ability: Interacts with other agents or humans.

Why PEAS is Needed: PEAS provides a structured framework to fully specify what an agent must do and what it has to work with. Without clearly defining PEAS, you cannot design an effective agent. It answers: What does success look like? What is the world like? How does the agent act? What can it sense?

PEAS Component	Taxi Driver Agent	Automated Manufacturing Robot
Performance Measure	Safe driving, speed, traffic law compliance, passenger comfort, maximize earnings	Number of parts assembled correctly, speed, minimal defects, safety, machine uptime
Environment	Roads, traffic lights, other vehicles, pedestrians, weather, passengers, GPS map	Factory floor, conveyor belt, raw materials, other robots, assembly line
Actuators	Steering wheel, accelerator, brake, horn, turn signals, door locks	Robotic arms, welders, grippers, drills, conveyor controls, LED displays
Sensors	Cameras, GPS, speedometer, radar/lidar, microphone, fuel gauge	Cameras, infrared sensors, weight sensors, barcode scanners, torque sensors, encoders

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Q8. List down the characteristics of well-defined problem. What is a rational agent? Justify 'System that thinks like humans' and 'System that acts like humans' with practical examples. [1+6] (2081 Ashwin, 2074 Bhadra)

Answer: Rational Agent

Rational Agent: An agent that acts to MAXIMIZE its expected performance measure given what it has perceived (percept sequence) and what it knows. Rationality is about making the BEST possible decision given available information — it does NOT require perfection.

System that Thinks Like Humans (Cognitive Modeling): This approach requires understanding the actual cognitive mechanisms of human thought — through psychology and neuroscience. A machine that 'thinks like humans' replicates the decision-making process of the human mind.

- Example: Cognitive architectures like ACT-R (Carnegie Mellon) model human memory, attention, and problem-solving. Psychological studies show humans use heuristics — AI systems like prospect-theory models mimic this.

System that Acts Like Humans (Turing Test Approach): A machine that produces output indistinguishable from a human's output — regardless of HOW it achieves this.

- Example: ChatGPT and Claude can write essays, poems, and have conversations that are indistinguishable from human writing. ELIZA (1966) acted like a therapist and fooled users despite simple rules.

System that Thinks Rationally (Laws of Thought): Based on formal logic and correct reasoning. Aristotle's syllogisms formalized this: IF all humans are mortal AND Socrates is human, THEN Socrates is mortal.

- Example: Theorem provers, FOPL-based reasoning systems, PROLOG.

System that Acts Rationally (Rational Agent): The dominant modern approach. The agent takes actions that maximize its expected performance measure — not necessarily mimicking human behavior.

- Example: AlphaGo doesn't play chess like humans think, but it plays BETTER by finding optimal moves through tree search and neural network evaluation.

SECTION 4: Knowledge, Learning & Disadvantages of AI

Exam Ref: 2079 Ashwin Back, 2078 Baishakh, 2077 Chaitra, 2075 Baisakh, 2073 Bhadra

Q9. What is knowledge and learning? Why are they important? Explain the factors required to pass the Turing Test. [2+2+4] (2079 Ashwin Back)

Answer: Knowledge & Learning

Knowledge: Knowledge in AI refers to the organized collection of facts, rules, relationships, and heuristics that an intelligent system uses to reason about the world and make decisions. Knowledge enables the agent to UNDERSTAND its environment — not just react to raw data.

Types of Knowledge:

Type	Description	Example
Declarative	Facts about the world ('knowing that')	'Kathmandu is the capital of Nepal'
Procedural	How to do something ('knowing how')	Algorithm for sorting a list
Heuristic	Rules of thumb based on experience	'If it looks like a duck and quacks like a duck...'
Structural	Relationships between concepts	'Heart is a part of the circulatory system'
Meta-Knowledge	Knowledge about knowledge	'I don't know the answer to this question'

Learning: Learning is the process by which an agent ACQUIRES new knowledge or skills, or IMPROVES existing ones, through experience. A learning agent gets better at its tasks without being explicitly reprogrammed.

Importance of Knowledge & Learning:

- Knowledge allows the agent to make informed, context-aware decisions.
- Learning allows the agent to adapt to new, unanticipated situations.
- Together they enable true autonomy — the agent doesn't need human guidance for every situation.

6 Factors Required to Pass the Turing Test:

Factor	Why Needed	Related AI Field
Natural Language Processing	To understand and respond in human language	NLP, Linguistics
Knowledge Representation	To store and reason about world knowledge	KR, Logic, Expert Systems
Automated Reasoning	To draw conclusions and answer questions	Theorem proving, Logic
Machine Learning	To learn and adapt to new conversational patterns	ML, Deep Learning
Computer Vision (Total TT)	To perceive and interpret visual information	CV, Image Processing
Robotics (Total TT)	To physically interact with the real world	Robotics, Control

Q10. List down disadvantages of AI. What is the importance of the Turing Test in Artificial Intelligence? What are the applications of AI? Explain. **[2+2+4]** (2078 Baishakh)

Answer: Disadvantages of AI

Disadvantage	Explanation
High Cost	AI development, hardware (GPUs), and training require enormous financial investment
Unemployment	AI automates jobs — displacing workers in manufacturing, data entry, customer service
Lack of Common Sense	AI systems fail in situations requiring basic human common sense and judgment
Data Dependency	AI requires massive amounts of quality data — poor data leads to poor AI
Bias and Fairness	AI can learn and amplify human biases present in training data (racial, gender bias)
Ethical Issues	Autonomous weapons, privacy violations, deepfakes, surveillance concerns
No Creativity	AI cannot truly create original ideas — it recombines existing patterns
Security Risks	AI systems can be attacked (adversarial attacks, data poisoning)
Over-reliance	Humans may become over-dependent on AI, losing critical thinking skills
Black Box Problem	Deep learning models are uninterpretable — hard to understand WHY they decide

Importance of Turing Test: The Turing Test is important because: (1) It provides an operational definition of intelligence — avoiding philosophical debates. (2) It sets a benchmark for AI progress. (3) It spurred development in NLP, knowledge representation, and reasoning. (4) It highlights the gap between current AI and human-level intelligence.

QUICK REVISION: Chapter 1 — Key Points Summary

Topic	Key Points	Expected Marks
Definition of AI	4 definitions: think/act like humans, think/act rationally; AI is about rational agents	2-4
Turing Test	Human judge, text conversation, cannot distinguish → machine passed	2-4
Types of Agents	Simple Reflex → Model → Goal → Utility → Learning (5 types with examples)	4-6
PEAS Framework	Performance, Environment, Actuators, Sensors — apply to Taxi/Robot/Doctor	4-6
AI vs Human Intelligence	Speed, memory, creativity, emotion, adaptability — AI won't fully replace humans	3-5
History of AI	1943 neuron → 1950 Turing → 1956 Dartmouth → 1997 Deep Blue → 2016 AlphaGo	2-4
AI Applications	Healthcare, NLP, vision, games, robotics, finance, education	2-4
Related Fields	Math, CS, Psychology, Neuroscience, Philosophy, Linguistics, Statistics	2-3
Knowledge & Learning	Declarative/Procedural/Heuristic knowledge; 6 Turing Test factors	2-4

Best of Luck! | TU IOE BCT | CT 653 AI | Chapter 1: Introduction